

The Periodic Table as ξ -Geometry:
Radial Mass Ladder and Angular Shell Structure
from a Common Origin

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Correction notes

Corrections from the calculation check (30 Sept. 2026). The following places are corrected or annotated in the text; each carries a dated note with the previous value:

- Rydberg energy: $m_e \alpha^2 / 2 = 13.5984 \rightarrow 13.6057$ eV (formula and table); 13.5984 eV is the ionisation energy of hydrogen with the reduced mass.
- Mass ladder $m_n \sim \xi^{-n} m_P \rightarrow \xi^n m_P$ (all places; the document's own lepton values $\xi^{+5.775} m_P$ etc.); CMB rung $\xi^4 \approx 10^{-16} \rightarrow \xi^{8.18} \approx 1.9 \times 10^{-32}$ (both hierarchy tables); Tab. A.2: noble-gas column Ne/Ar/Kr $\rightarrow$ Ne, Ar / Kr, Xe / Rn, Og.
- (2 Oct. 2026) Notation f : correction function $f(n)$ — not a frequency; note in the text (Doc. 190, R147).

Chapter A

The Periodic Table as ξ -Geometry

Abstract

The periodic table of the elements and the particle-mass ladder of T0 theory share the same geometric quantum number n , but in different roles. Particle masses follow a *radial* exponential scaling $m_n \sim \xi^n \cdot m_p$, while the shell structure of the periodic table obeys an *angular* quadratic degeneracy $g_n = 2n^2$. Both arise from the three-dimensional lattice geometry of T0 theory: ξ is the packing deficit of the 3D lattice (radial direction), $2n^2$ is the degeneracy of the n -th spherical shell in the same lattice (angular direction). The Rydberg energy lies exactly on rung 7 of the ξ -power ladder: $R_\infty = \frac{m_e \alpha^2}{2} \approx \xi^{6.956} \cdot E_p$. The duality principle $E_n \cdot g_n = 2R_\infty = \text{const}$ connects the $1/n^2$ energy ladder with the n^2 filling ladder and is a geometric equipartition. The periodic table is the angular cross-section of the ξ -geometry at rung $\xi^{\sim 7}$ — analogously, the lepton masses are the radial cross-section at rungs $\xi^{4.9}$ – $\xi^{5.8}$.

A.1 Introduction and Problem Statement

T0 theory (Time-Mass Duality / FFGFT) derives all physical constants from the geometric parameter

$$\xi = \frac{4}{30\,000} = 1.3 \times 10^{-4} \quad (\text{A.1})$$

[1, 3, 4]. The particle masses of the Standard Model organise themselves on an exponential ladder

$$m_n \sim \xi^n \cdot m_P, \quad n = 1, 2, 3, \dots \quad (\text{A.2})$$

where n is the T0 quantum number and m_P is the Planck mass.

The periodic table of the elements, on the other hand, shows a very different pattern: shell capacities follow

$$g_n = 2n^2 = 2, 8, 18, 32, 50, \dots \quad (\text{A.3})$$

and binding energies scale as $1/n^2$. At first glance, particle masses and the periodic table appear unrelated.

Gemini's observation (2026): the periodic table lies "one level higher" than the particle masses — both have quantum numbers, but what common geometric origin connects their patterns?

The answer of this document: *T0 geometry has two axes — a radial one (ξ^n scaling) and an angular one ($2n^2$ degeneracy), both arising from the three-dimensional lattice structure with packing deficit ξ .*

A.2 The ξ -Power Ladder and Its Rungs

Previous documents in the T0 series have built up the ξ -power ladder step by step. Table A.1 shows the current state including the new atomic rung.

Table A.1: Complete ξ -power ladder (as of March 2026)

Rung	ξ power	Physical content	Document
0	$\xi^0 = 1$	Planck scale, $L_0 = \xi \cdot \ell_P$	Doc. 009
1/2	$\xi^{1/2} \approx 10^{-2}$	$\alpha_W, m_e/m_{\mu}, \text{PPLN } \Delta n$	Doc. 041, 167
1	$\xi^1 \approx 10^{-4}$	$\alpha_{\text{EM}} \sim \xi \cdot E_0^2$	Doc. 011
23/4	$\xi^{5.775}$	Electron mass m_e/m_P	Doc. 056
~ 7	$\xi^{6.956}$	Rydberg energy R_∞, periodic table	this doc.
~ 8.18	$\xi^{8.18} \approx 1.9 \times 10^{-32}$	CMB temperature $k_B T_{\text{CMB}}/E_P$	Doc. 166
10	$\xi^{10} \approx 10^{-43}$	Hubble constant H_0	Doc. 165

(Corrected on 30 Sept. 2026: CMB row previously rung 4, $\xi^4 \approx 10^{-16} - k_B T_{\text{CMB}}/E_P = 2.349 \times 10^{-4} \text{ eV} / 1.2209 \times 10^{28} \text{ eV} = 1.92 \times 10^{-32} = \xi^{8.18}$, whereas $\xi^4 = 3.2 \times 10^{-16}$; Doc. 166 contains no ξ^4 assignment for T_{CMB} . Row moved to match the order of exponents.)

The Rydberg energy lies between rungs 6 and 7:

$$\frac{R_\infty}{E_P} = \frac{m_e}{m_P} \cdot \frac{\alpha^2}{2} = \xi^{5.775} \cdot \xi^{1.181} = \xi^{6.956} \approx \xi^7 \quad (\text{A.4})$$

The exponent is the sum of two independent T0 quantities: the electron mass (rung 5.775, Doc. 056) and the fine-structure constant (rung 1.181, Doc. 011).

A.3 Two Axes of ξ -Geometry

Radial Axis: Exponential Mass Ladder

Particle masses organise themselves along the *radial* axis of the T0 lattice. The fundamental relation is [1]:

$$m_n \sim \xi^n \cdot m_P, \quad n = 1, 2, 3, \dots \quad (\text{A.5})$$

Each step on this ladder ($n \rightarrow n + 1$) divides the mass by $\xi^{-1} = 7500$. (*Corrected on 30 Sept. 2026: previously $m_n \sim \xi^{-n} \cdot m_P$ (here and at all further places) and "multiplies the mass by 7500" – $\xi^{-n}m_P = 7500^n m_P$ would lie above the Planck mass; the lepton values below read $\xi^{+5.775}m_P$ etc., since $\ln(m_e/m_P)/\ln \xi = \ln(4.19 \times 10^{-23})/\ln(1.333 \times 10^{-4}) = +5.775$.) The leptons occupy the rungs (Doc. 056, 122):*

$$m_e : \quad \xi^{5.775} \cdot m_P \quad (\text{A.6})$$

$$m_\mu : \quad \xi^{5.177} \cdot m_P \quad (\text{A.7})$$

$$m_\tau : \quad \xi^{4.861} \cdot m_P \quad (\text{A.8})$$

The mass ratio $m_e/m_\mu \propto \xi^{1/2}$ (Doc. 041) lies exactly on the electroweak exponent — the half-rung of the ξ -ladder.

Angular Axis: Quadratic Degeneracy Ladder

The *angular* axis of the same lattice determines the degeneracy of spherical shells. In 3D space, the n -th principal shell has:

$$g_n = 2 \sum_{l=0}^{n-1} (2l+1) = 2n^2 \quad (\text{A.9})$$

The factor 2 comes from spin $m_s = \pm \frac{1}{2}$; the sum is pure angular-momentum algebra in 3D space. Table A.2 shows the first four shells.

Table A.2: Shell structure and the periodic table

n	$g_n = 2n^2$	Period length	Noble gas Z	Period number
1	2	2	He (2)	1
2	8	8	Ne (10), Ar (18)	2, 3
3	18	18	Kr (36), Xe (54)	4, 5
4	32	32	Rn (86), Og (118)	6, 7

(*Corrected on 30 Sept. 2026: noble-gas column previously Ne (10) / Ar (18) / Kr (36) for $n = 2, 3, 4$ – the periods of length 8 (2, 3) end at Ne (10) and Ar (18), those of length 18 (4, 5) at Kr (36) and Xe (54), those of length 32 (6, 7) at Rn (86) and Og (118); the column was shifted by one row from $n = 3$ on.*)

The Connection through Quantum Number n

The same quantum number n appears in both systems, but with different functional roles:

$$\text{Radial (masses): } m_n \sim \xi^n \cdot m_P \quad (\text{exponential}) \quad (\text{A.10})$$

$$\text{Angular (shells): } g_n = 2n^2 \quad (\text{quadratic}) \quad (\text{A.11})$$

This is not coincidental. In T0 lattice geometry, n corresponds to the *discrete radius in the lattice*:

- Radially: n steps outward from the lattice $\Rightarrow$ mass scales as ξ^n (geometric sequence)
- Angularly: the lattice points on the sphere of radius n carry degeneracy $2n^2$ (combinatorial sequence)

A.4 The Duality Principle

Energy Ladder and Filling Ladder Are Dual

On the atomic scale (hydrogen-like atoms):

$$\text{Energy ladder: } E_n = -\frac{R_\infty}{n^2} \quad (\text{falls as } 1/n^2) \quad (\text{A.12})$$

$$\text{Filling ladder: } g_n = 2n^2 \quad (\text{grows as } n^2) \quad (\text{A.13})$$

Their product is exactly constant:

$$E_n \cdot g_n = \frac{R_\infty}{n^2} \cdot 2n^2 = 2R_\infty = \text{const} \quad (\text{A.14})$$

This is geometric *equipartition*: every shell contributes the same total energy $2R_\infty$ to the Rydberg scale, regardless of n . The $1/n^2$ energy ladder and the n^2 filling ladder are *inversely dual*.

Comparison with the Mass Ladder

Table A.3 summarises all three ladders.

Table A.3: Three ladders — one geometric origin

System	Ladder	Function of n	Type
Particle masses	$m_n \sim \xi^n \cdot m_P$	exponential	radial
Shell energy	$E_n = R_\infty/n^2$	harmonic	angular
Shell filling	$g_n = 2n^2$	quadratic	angular
Duality: $E_n \cdot g_n = 2R_\infty = \text{const}$			

A.5 The Role of Fractal Dimension

In T0 (Doc. 009, 133):

$$D_f = 3 - \xi = 2.999867 \quad (\text{A.15})$$

Space dimension is almost, but not exactly, 3.

In the limit $D_f \rightarrow 3$ ($\xi \rightarrow 0$):

- $g_n = 2n^2$ is *exact* (pure spherical symmetry in the $\mathbb{Z}^3$ lattice)
- the shell structure of the periodic table is *exact*

For $D_f = 3 - \xi < 3$ there is a minimal correction:

$$\delta g_n \sim \xi \cdot n^2 \quad (\text{A.16})$$

For $n = 2$: $\delta g_2 \approx \xi \cdot 4 \approx 5 \times 10^{-4}$ (entirely negligible).

The periodic table is therefore exact because $D_f \approx 3$ to four decimal places. The tiny deviation $\xi \approx 1.3 \times 10^{-4}$ makes space "almost-integer-dimensional" — and precisely this deviation generates all the physics of particle masses and fundamental constants.

A.6 Numerical Verification

Rydberg Energy on the ξ -Ladder

$$R_\infty = \frac{m_e \alpha^2}{2} = 13.6057 \text{ eV} \quad (\text{A.17})$$

$$\frac{R_\infty}{E_P} = \frac{m_e}{m_P} \cdot \frac{\alpha^2}{2} = \xi^{5.775} \cdot \xi^{1.181} = \xi^{6.956} \quad (\text{A.18})$$

(Corrected on 30 Sept. 2026: previously 13.5984 eV – $m_e \alpha^2 / 2 = 510\,998.95 \times (1/137.036)^2 / 2 = 13.6057 \text{ eV}$; 13.5984 eV is the hydrogen ionisation energy with the reduced mass. Likewise in the table; the exponent 6.956 is unchanged.)

Table A.4: Rydberg energy: T0 derivation vs. measurement

Quantity	Value	ξ -exponent
m_e/m_P	4.185×10^{-23}	5.775
$\alpha^2/2$	2.663×10^{-5}	1.181
R_∞/E_P	1.114×10^{-27}	6.956
R_∞ (T0, approx.)	$\approx 9.1 \text{ eV } (\xi^7)$	7.000
R_∞ (measured)	13.606 eV	6.956

Duality Relation: Numerical Verification

$$E_1 \cdot g_1 = 13.60 \text{ eV} \cdot 2 = 27.20 \text{ eV} = 2R_\infty \quad (\text{A.19})$$

$$E_2 \cdot g_2 = 3.40 \text{ eV} \cdot 8 = 27.20 \text{ eV} = 2R_\infty \quad (\text{A.20})$$

$$E_3 \cdot g_3 = 1.51 \text{ eV} \cdot 18 = 27.18 \text{ eV} \approx 2R_\infty \quad (\text{A.21})$$

$$E_4 \cdot g_4 = 0.85 \text{ eV} \cdot 32 = 27.20 \text{ eV} = 2R_\infty \quad (\text{A.22})$$

The duality relation (A.14) holds exactly for all n .

A.7 The Periodic Table in the T0 Hierarchy

ξ -rung	Physics	Axis
$\xi^0 = 1$	Planck scale ($L_0 = \xi \cdot \ell_P$)	—
$\xi^{1/2}$	Electroweak: $\alpha_W, m_e/m_\mu$	radial
ξ^1	α_{EM} , EM coupling	radial
$\xi^{5.8}$	Particle masses (leptons, quarks)	radial
$\xi^{6.956}$	Rydberg R_∞, periodic table	both
$\xi^{8.18}$	CMB temperature $k_B T_{CMB}/E_P$	radial
ξ^{10}	Hubble constant H_0	radial

Figure A.1: Position of the periodic table in the complete ξ -hierarchy

(Corrected on 30 Sept. 2026: CMB row previously $\xi^4 - k_B T_{CMB}/E_P = 1.92 \times 10^{-32} = \xi^{8.18}$; row moved to match the order of exponents.)

The special feature of the atomic rung ξ^7 : here *both* axes act. The radial axis determines the energy scale R_∞ ; the angular axis determines the pattern $2n^2$. The periodic table is the visible result of both axes acting together.

A.8 Predictions and Open Questions

Angular Corrections from $D_f < 3$

The fractal dimension $D_f = 3 - \xi$ leads to minimal corrections of the exact $2n^2$ rule:

$$g_n^{(T0)} = 2n^2 \cdot (1 - \xi \cdot f(n)) \quad (\text{A.23})$$

where $f(n)$ is a dimensionless function of n (still to be derived). For $n \leq 4$ the correction is $< 10^{-3}$ and not measurable.

(Note of 2 Oct. 2026 on notation: here $f(n)$ is the dimensionless correction function of the $2n^2$ rule, not a frequency and not the base winding number $f = 1/\xi$. Cf. Doc. 190, R147.)

Molecular Binding Energies

Chemical binding energies lie in the range 1–10 eV, hence also on rung ξ^7 . It remains to be checked whether specific binding energies lie on rational multiples of $\xi^{7/2} \cdot R_\infty$.

Heavy Elements and Relativistic Corrections

For $Z \gg 1$ relativistic effects become important. In T0 these are connected with the $\xi^{1/2}$ electroweak rung. A systematic analysis of ionisation energies of heavy elements ($Z > 54$) might reveal ξ -matches at the half-rung $\xi^{7-1/2} = \xi^{13/2}$.

A.9 Summary

1. **Energy scale:** $R_\infty/E_P = \xi^{6.956} \approx \xi^7$. The Rydberg energy — and hence all of atomic physics — lies on rung 7 of the ξ -power ladder.
2. **Radial axis:** Particle masses follow $m_n \sim \xi^n \cdot m_P$. The exponent n is the discrete lattice radius in the radial ξ -direction.
3. **Angular axis:** Shell capacities follow $g_n = 2n^2$. The index n is the same discrete lattice radius, now acting as a shell index in the angular 3D-sphere direction.
4. **Duality principle:** $E_n \cdot g_n = 2R_\infty = \text{const.}$ The $1/n^2$ energy ladder and the n^2 filling ladder are inversely dual. Every shell contributes the same total energy $2R_\infty$ to the ξ^7 scale.
5. **Fractal dimension:** $D_f = 3 - \xi \approx 3$ makes $2n^2$ exact (to $\sim 10^{-4}$). The periodic table is exact because space is almost-integer-dimensional.
6. **Common origin:** Particle masses and the periodic table arise from the same 3D lattice geometry with packing deficit ξ . The periodic table is the *angular cross-section* of ξ -geometry at rung ξ^7 ; the lepton masses are the *radial cross-section* at rungs $\xi^{4.9} - \xi^{5.8}$.

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